

System Design Review

Independent SDR / PDR assessment of the Boaty Mk1 system documentation baseline, with findings, recommendations and the plan to freeze

Document	BOATY-SDR-001
Issue	Issue A (review report)
Date	30 September 2026
Review type	System Design Review / Preliminary Design Review (gate 1 of 4: SDR, CDR, TRR, ORR)
Recommendation	Conditional GO: freeze after closing 9 Major and 6 Minor RIDs
Findings	9 Major, 6 Minor, 7 Observations carried to a named gate

Document control

Issue	Date	Change	By
A	30 September 2026	First issue: review of the documentation baseline as at commit 88b1cd7.	Claude (as reviewer)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

How to use this report: section 2 gives the decision. Section 6 lists every review item discrepancy (RID) with its evidence and recommended closure. Section 8 is the plan to freeze. Close a RID by recording the closure evidence in the RID log (section 8.3); the freeze happens when every Major and Minor RID is closed or dispositioned by the owner.

Contents

1. Purpose, scope and reviewer statement	4
1.1 Scope	4
1.2 Method	4
1.3 Independence statement	4
2. Summary and recommended decision	5
2.1 What is good	5
2.2 What must change before freeze	5
3. Documents reviewed	6
4. Entry criteria assessment	6
5. Compliance assessment	7
5.1 Requirements	7
5.2 Architecture and allocation	7
5.3 Interfaces	7
5.4 Verification status by stage	7
5.5 Safety and risk	7
5.6 Budgets	7
5.7 Configuration management	7
6. Review item discrepancies	9
6.1 Severity definitions	9
6.2 RID summary	9
6.3 RID detail	10
7. Items carried to later gates	15
8. Gap-closure plan and freeze	16
8.1 Work packages	16
8.2 Freeze (SDR exit) criteria	16

8.3 RID log	16
8.4 What CDR will expect	17
9. Decision record	18
Appendix A. SIM-stage Must requirements without a named test	19

1. Purpose, scope and reviewer statement

This is the first of the four project gates agreed with the owner. SDR/PDR asks one question: is the system design, as documented, complete, consistent and credible enough to freeze and start detailed design? Detailed design here means hardware drawings, the build, and the remaining software. The gate comes before any Mk1 hardware is bought, so evidence is analysis, inspection and simulation.

1.1 Scope

- All controlled documents in docs/ as built at commit 88b1cd7 (list in section 3).
- The parameter baseline (software/params) and the site file, as configuration items that the documents claim.
- The verification evidence in software/results: SITL records (46), unit evidence, and the live Claude evaluation (41 cases).
- Out of scope: detailed code review (that belongs to CDR), hardware (none exists) and the lake site (not surveyed).

1.2 Method

- Read every document end to end against the SRS as the top-level contract.
- Cross-checked each document's stated parents, issues and facts against the others (configuration consistency).
- Traced Must requirements into the design (ADD allocation, SSS derived requirements), and on into verification (catalogue, results), using the source data rather than the rendered PDFs, so the counts here are exact.
- Checked the claims the documents make about configuration (parameters, site, tests) against the repository itself.
- Graded findings as Major, Minor or Observation (section 6.1).

1.3 Independence statement

The owner asked for an independent review. The reviewer is the same assistant that drafted most of these documents, which is not true independence. To offset this, the review worked from the source data and the repository rather than the prose. It looked first for where the documents contradict the code. **Recommendation:** the owner, as a practising engineer, should personally read the Major RIDs and the safety sections (FMEA, failsafes) before signing the freeze. A second person (another engineer, or another AI session given only the documents) would strengthen it further.

2. Summary and recommended decision

Recommendation: Conditional GO. The architecture is sound, well reasoned and unusually well evidenced for this stage. 25 of 37 catalogued simulator scenarios already pass on the real software. But the documentation is **not yet a baseline that can be frozen**. One Must safety requirement is not met as configured (RTL crossing an exclusion). The failsafe verification the SRS demands is incomplete. Two requirements contradict the design. The wildlife rule conflicts with the product goal. And the documents disagree with each other about which issues they depend on. Close the 9 Major and 6 Minor RIDs, rebuild and tag, then freeze.

2.1 What is good

- **Safety architecture.** Every safety function (fence, failsafes, RTL, STOP) sits in the flight controller. Python and Claude can only propose. The validator is the only way to create a mission the helm will accept, and this is enforced in code, not just stated.
- **Documents built from data.** Requirements, allocation, parameters and test results are generated from one source. Builds fail on untraced requirements, unknown references and a BOM over the cap. That is better configuration discipline than many funded projects manage.
- **Evidence ahead of phase.** 43 SITL records pass. There are 295 unit tests, and property-based validator fuzzing. The live Claude evaluation got 41/41 with every must-decline case declined (14/14). The simulator has already found and fixed real design faults (SC-06, the parameter read-back, short-trip planning). They are fed back into the FMEA.
- **Operations taken seriously.** The operations manual, crew card and checklist are traced to the FMEA's procedural actions. The same checklist drives Mission Control.

2.2 What must change before freeze

RID	Issue	Closure in one line
RID-01	RTL is not fence-aware (NAV-007, HLM-D19)	Configure ArduPilot object avoidance with the Dijkstra planner around fences (OA_TYPE = 2 and its margins) in the baseline.
RID-02	SWE-005 not met: FS-011, 012 and 013 have no automated scenario	Build SC-11, SC-12 and SC-13 in the existing SITL harness.
RID-03	Residual risk above threshold with no disposition (FM-02)	Build SC-20 (20-50 m jump for 2 s near the fence) and re-rate FM-02 on the result.
RID-04	Photographing ducks conflicts with OPS-005 wildlife rule	Owner decision, recorded as a CR.
RID-05	Event vocabulary is an uncontrolled interface	Define the event vocabulary once in code (one module of constants used by B4-B7 and C-side parsing).
RID-06	Two requirements contradict the agreed design	Raise CR-06 to amend both.
RID-07	No requirement-level verification matrix (VCRM)	Generate a VCRM from source: SRS requirement → SSS derived requirement(s) → test IDs (catalogue and pytest node IDs) → latest result.
RID-08	Document set is not a consistent baseline	Create one baseline register in code (docs/common/baseline.py: document → issue → date).
RID-09	Software baseline is not identifiable	Pin ArduPilot to one commit everywhere (KCL, README, build script) and re-archive the sources if needed.

3. Documents reviewed

Document	ID	Issue	Reviewer's note
Concept Selection Report	-	v1.1	Superseded in detail by the SRS and ADD; kept as the rationale record. No action.
System Requirements Spec.	BOATY-SRS-001	F	184 requirements. Stale text (RID-08); two contradictions (RID-06); OPS-005 (RID-04).
Architecture Design Doc.	BOATY-ADD-001	F	Sound; stale drivers, cost and references (RID-08); security assumption missing (RID-15).
Interface Control Doc.	BOATY-ICD-001	F (PDF stale)	22 interfaces. PDF not rebuilt; IF-04 vocabulary (RID-05); IF-11 text (RID-06, RID-14).
Subsystem specs (8)	BOATY-SSS-*	A-E	243 derived requirements. HUL and REC Issue A with stale parents; hard-coded references (RID-08).
Design FMEA	BOATY-FMEA-001	E	58 failure modes, 28 actions. FM-02 (RID-03), FM-13 (RID-01), FM-44 (RID-04), rules (RID-13).
Operations Manual	BOATY-OPS-001	C	Good. Needs the wildlife and data statements (RID-04, RID-14).
Key Component List	BOATY-KCL-001	B	BOM £181 of £185. ArduPilot commit mismatch (RID-09); basis stale (RID-08).
Software and results	software/	commit 88b1cd7	Not a document, but the documents cite it. No tag, no lock, no CI (RID-09, RID-10).
Not provided and not needed at this gate: detailed drawings, the wiring diagram, the software design description and the test procedures. They are CDR deliverables (section 8.4).			

4. Entry criteria assessment

#	Entry criterion	Status	Evidence / RID
E1	Stakeholder needs and ConOps agreed	Met	SRS §1-2; STK-01 to STK-08, with owner decisions logged.
E2	System requirements complete, with priority, method and stage	Met with RIDs	184 requirements; 2 of 8 TBDs open (site-dependent). RID-04, RID-06.
E3	Architecture selected with trade studies	Met	ADD design-space exploration, weighted candidates and sensitivity.
E4	Every requirement allocated to a subsystem	Met	The ADD and SSS builds fail on unallocated requirements.
E5	Interfaces identified and defined	Partly met	22 interfaces defined; event vocabulary uncontrolled (RID-05); TBCs open (OBS-03).
E6	Subsystem requirements derived and traced	Met	243 derived requirements; trace checked by the build.
E7	Budgets (cost, mass, power) with margins	Partly met	Cost £181/£185; power.csv missing (RID-12); the numbers are estimates (OBS-04).
E8	Hazards and failure modes analysed; high risks actioned	Partly met	58 FMs; FM-02 has no disposition (RID-03); FM-13 relies on an unconfigured control (RID-01).
E9	Verification approach defined per requirement	Partly met	Method and stage on every requirement, but no VCRM (RID-07).
E10	Risky technology de-risked	Met	ArduPilot SITL and the live LLM evaluation already run on the real software.
E11	Documents under configuration control	Not met	Cross-references inconsistent; no tags (RID-08, RID-09).

5. Compliance assessment

5.1 Requirements

The SRS holds 184 requirements: 137 Must, 40 Should and 7 Could. Each has a verification method, a stage and a trace to a stakeholder need, and the build rejects any that do not. By stage: SIM 76, BENCH 62, LAKE 24, POOL 22. The requirements are mostly singular, testable and free of design. The exceptions are those in RID-06, and HLM-D19 which names a mechanism.

TBDs: 6 of 8 closed. The two open ones (TBD-02 site and TBD-03 permission) don't affect the system design. They are carried to ORR (OBS-01).

5.2 Architecture and allocation

The allocation is complete by construction. The division of authority is the design's strongest feature: the flight controller alone is safety authority, Python can only request HOLD, and Claude can only propose. It is enforced in the code in three ways: ValidatedMission can only be created by the validator, GuardedHelm refuses motion commands while a foreign GCS is present, and parameter read-back is checked against the baseline. The accepted common-cause risk (fence and failsafes in one flight controller) is stated honestly in the FMEA residual register, with a Mk2 path (SAF-005).

5.3 Interfaces

The ICD defines all 22 interfaces from the ADD register, with owners, timing and error handling. The weakness is what sits inside an interface. The STATUSTEXT vocabulary (RID-05) and the IF-11 payload (RID-06, RID-14) are defined by code rather than by the ICD.

5.4 Verification status by stage

Stage	SRS reqs	Catalogue tests	Status at SDR
SIM	76	37	25 pass, 12 not built (SC-08, SC-11, SC-12, SC-13, SC-20, SC-21, SC-22, SC-23, SC-27, SC-34, SC-36, SC-39). No failures.
BENCH	62	7	Planned; needs hardware (CDR/TRR).
Rigs L1-L3	-	26	Planned; needs hardware.
POOL	22	1	Planned (TRR).
LAKE	24	-	Planned (ORR onwards).

SITL run: 43 passed, 2 known gaps (expected failures for later slices), 1 known gap not seen. Live NLI evaluation: 41/41, p95 6.9 s. Failsafes: 13 of 13 FS requirements are named by a catalogue test, but only those with a built scenario count for SWE-005 (RID-02).

5.5 Safety and risk

The FMEA has 58 failure modes. 12 are at severity ≥ 9 , and all of those have a test. 18 met the RPN ≥ 100 rule before actions and 2 after (FM-02, FM-43). 4 residual risks are formally accepted (FM-43, FM-31, FM-23, Common cause). The FMEA is strengthened by simulator findings (FM-49 to FM-58). Its weaknesses are the unconfigured RTL control (RID-01), the open FM-02 (RID-03), the wildlife rating (RID-04) and one missing mode (RID-13).

5.6 Budgets

Budget	Value	Limit / margin	Assessment
Cost (BOM)	£181	£185 cap; £4 margin	Compliant but thin; estimated prices (RID-12).
Power	70 W limit	PWR-D17 $\geq 30\%$ margin	Analysis in KCL; budget file missing (RID-12).
Mass / buoyancy	SSS-HUL	Reserve buoyancy	Estimated; CDR to confirm with a weighed build.
API cost	\$0.31 per 41 plans	Pennies per trip	Compliant.

5.7 Configuration management

Generating documents from source is excellent practice. But each document hard-codes its parents' issue letters, so they drift (RID-08). The software has no tags, lock file or CI (RID-09, RID-10). A freeze needs one identifier that names a consistent set: documents, parameters, site and code.

6. Review item discrepancies

6.1 Severity definitions

Severity	Meaning	Before freeze?
Major	The baseline fails a Must requirement, is internally contradictory, or cannot be identified. Would mislead detailed design.	Close
Minor	A gap or error that is local and cheap to fix, and that does not change the architecture.	Close or owner disposition
Obs.	Expected to be open at this phase. Carried to the named gate.	Carry

6.2 RID summary

RID	Sev.	Area	Title
RID-01	Major	Safety	RTL is not fence-aware (NAV-007, HLM-D19)
RID-02	Major	Verification	SWE-005 not met: FS-011, 012 and 013 have no automated scenario
RID-03	Major	Safety	Residual risk above threshold with no disposition (FM-02)
RID-04	Major	Safety / stakeholder	Photographing ducks conflicts with OPS-005 wildlife rule
RID-05	Major	Interfaces	Event vocabulary is an uncontrolled interface
RID-06	Major	Requirements	Two requirements contradict the agreed design
RID-07	Major	Verification	No requirement-level verification matrix (VCRM)
RID-08	Major	Configuration	Document set is not a consistent baseline
RID-09	Major	Configuration	Software baseline is not identifiable
RID-10	Minor	Configuration	No CI, lint or type checking
RID-11	Minor	Safety	HDOP arming gate not configured
RID-12	Minor	Budgets	Power budget file missing; cost margin thin
RID-13	Minor	Safety	FMEA rule text and the build check disagree; missing failure mode
RID-14	Minor	Privacy	Child's voice transcripts go to a cloud service without a stated policy
RID-15	Minor	Security	Web UI security assumptions not recorded
OBS-01	Obs.	Operations	Site and permission open (TBD-02, TBD-03)
OBS-02	Obs.	Verification	Hardware verification items open
OBS-03	Obs.	Interfaces	ICD TBC items open
OBS-04	Obs.	Budgets	Performance numbers are estimates
OBS-05	Obs.	Performance	Planning latency not measured on the real link
OBS-06	Obs.	Scope	Features not yet implemented
OBS-07	Obs.	Verification	Remaining FMEA scenarios not built

6.3 RID detail

RID-01	Major	RTL is not fence-aware (NAV-007, HLM-D19)
Area	Safety	Documents: SRS NAV-007; SSS-HLM HLM-D19; FMEA FM-13; KCL; software/params
Finding	NAV-007 (Must) needs RTL to reach home without entering an exclusion zone. HLM-D19 allocates this to 'fence-aware path planning' in the helm. The parameter baseline has no OA_* parameters , so a failsafe or adult RTL steers a straight line home and can cross the island exclusion. FM-13 (S8) is rated D3 on the strength of this control and of SC-27, which is not built. This is the one gap where the design, as configured, does not meet a Must safety requirement.	
Recommend	Configure ArduPilot object avoidance with the Dijkstra planner around fences (OA_TYPE = 2 and its margins) in the baseline. Build SC-27 (breach on the far side of the island, then RTL) and SITL-verify a clean path. If SITL shows it is unreliable with this hull, change NAV-007 instead: put it in the site rules that no exclusion may lie between home and any mission area (a site-linter check). Then re-rate FM-13.	
Close with	Parameter file diff, SC-27 pass record, FM-13 re-rated	
RID-02	Major	SWE-005 not met: FS-011, 012 and 013 have no automated scenario
Area	Verification	Documents: SRS SWE-005, FS-008/011/012/013; SSS-SIM catalogue
Finding	SWE-005 (Must, SIM stage) requires an automated simulation scenario for every failsafe FS-001 to FS-013. The catalogue lists SC-11 (GNSS loss during battery RTL), SC-12 (log and announce audit) and SC-13 (single-fault sweep), but none is built. SC-08 (FS-008, helm signal loss to the ESCs) is also not built. The SSS moves it to rig L2, but the SRS still asks for a simulation. FS-013 (no single failure leaves the fence under power) is the system's top safety claim and has no automated evidence.	
Recommend	Build SC-11, SC-12 and SC-13 in the existing SITL harness. SC-13 can be a parametrised sweep over the SC-01 to SC-05 fault injectors × mission phase. For FS-008, either write an SRS deviation (verified on rig L2 by test, as SITL cannot model the ESC input) or build a SITL analogue (servo output frozen or zeroed).	
Close with	Three pass records in sitl_results.json; SRS deviation or SC-08 record	
RID-03	Major	Residual risk above threshold with no disposition (FM-02)
Area	Safety	Documents: FMEA FM-02, residual-risk register, SSS-SIM SC-20
Finding	18 failure modes met the RPN ≥ 100 rule before actions and 2 still do after them: FM-02 (108), FM-43 (126). FM-43 (swimmer near the boat) is formally accepted in the residual-risk register. FM-02 (GNSS position jump near the fence, S9) is neither accepted nor closed. Its post-action rating waits on SC-20, which is not built.	
Recommend	Build SC-20 (20-50 m jump for 2 s near the fence) and re-rate FM-02 on the result. If the rating stays ≥ 100, the owner must accept it in the residual-risk register with its rationale (e.g. EKF glitch rejection plus a 100 m fence set back from the shore).	
Close with	SC-20 record; FM-02 re-rated or residual entry signed	
RID-04	Major	Photographing ducks conflicts with OPS-005 wildlife rule
Area	Safety / stakeholder	Documents: SRS OPS-005, STK-06; site file; planner templates; FMEA FM-44; NLI evaluation set
Finding	The mission statement is 'explore the pond, take pictures of ducks'. OPS-005 (Must) says 'no mission shall aim to approach or follow wildlife' and makes known nesting areas exclusions. The stand-in site has a duck house that is both a 2 m exclusion and a photo landmark with a 6 m keep-out. The live evaluation planned visits to it. The 'Duck patrol' template makes wildlife stops a product feature. A floating duck house is a likely nesting site in the breeding season (roughly March to July). Disturbing nesting birds can be an offence. FM-44 (nesting area approached, O2) looks understated when the system is designed to go there.	
Recommend	Owner decision, recorded as a CR. (1) Reword OPS-005 so the system may photograph wildlife from a set stand-off but never chase it. (2) Set a nesting stand-off that the site linter enforces (suggest ≥ 15 m exclusion radius around nest structures, not a 6 m landmark keep-out). (3) Add a seasonal flag to site features, so landmarks are unavailable in the breeding season. (4) Rename or constrain 'Duck patrol'. (5) Re-rate FM-44 and add the rule to the operations manual and the NLI declines.	
Close with	CR approved; site schema and linter change; FM-44 re-rated	
RID-05	Major	Event vocabulary is an uncontrolled interface
Area	Interfaces	Documents: ICD IF-04; SSS-MCN MCN-D60; SSS-MCP B4-B7; software

Finding	The boat services tell Mission Control what they are doing through STATUSTEXT messages starting 'BOATY '. MCN-D60 (spoken reasons for a HOLD) parses the text of these messages. IF-04 mentions only the prefix. The actual vocabulary (e.g. the new B5 episodes, 'REPEATEDLY STUCK: HOLD', 'NO CONTROL') exists only in code. A wording change on the boat would silently break a verified requirement on shore.
Recommend	Define the event vocabulary once in code (one module of constants used by B4-B7 and C-side parsing). Generate an IF-04 table from it in the ICD build. Add a unit test that every event the services can emit is known to the parser. Add the B5 episode cap and its parameters to IF-04.
Close with	ICD IF-04 table; shared constants module; unit test

RID-06	Major	Two requirements contradict the agreed design
Area	Requirements	Documents: SRS NLI-003, MOD-005; ICD IF-11; ADD
Finding	NLI-003 requires the Claude request to include 'fence, exclusion zones, home ...'. IF-11 deliberately sends named places and a trip-time guide, not geometry (data minimisation; Claude never sees coordinates, and the validator owns geometry). MOD-005 says a finished AUTO mission 'shall enter RTL'. In the design, the last mission item is a return-to-launch that runs in AUTO. The mode reported stays AUTO until HOLD. Both designs are better than the requirement text, but a frozen baseline cannot hold requirements known to be false.	
Recommend	Raise CR-06 to amend both. NLI-003: 'the request shall include the site's named places, the mission schema and the time and energy limits, and shall not include coordinates'. MOD-005: 'the boat shall return home (RTL behaviour, in AUTO or RTL mode) and then HOLD'. Update the MOD table and ICD IF-11 wording to match.	
Close with	SRS issue with CR-06	

RID-07	Major	No requirement-level verification matrix (VCRM)
Area	Verification	Documents: SRS; SSS; SSS-SIM catalogue; software/results
Finding	The SRS has 184 requirements. 76 are due for verification at the SIM stage. 65 of those (53 Must) are not named in the trace of any catalogue test. Many are in fact verified by unit tests or by SSS-derived requirements, but nothing shows it requirement by requirement. The reviewer could not confirm SIM-stage compliance for these without reading test code. This is the main exit criterion of a PDR, and it is the document that CDR and TRR will need.	
Recommend	Generate a VCRM from source: SRS requirement → SSS derived requirement(s) → test IDs (catalogue and pytest node IDs) → latest result. Tag pytest tests with the requirement IDs they verify (a marker) so the matrix is built, not typed. The build should fail if a Must SIM-stage requirement has neither evidence nor a named future test. Appendix A lists the gaps to start from.	
Close with	VCRM appendix or document; build check	

RID-08	Major	Document set is not a consistent baseline
Area	Configuration	Documents: All documents
Finding	The documents cross-reference issues and facts that have since moved on. Examples found: • ICD PDF not rebuilt after the source change (shows parent ADD Issue E). References [1] still cite SRS D, ADD D and FMEA B. IF-11 says 'tool use' and 'keyring' (the design uses structured output and a 0600 key file). IF-21 says '106 passing'. The IF-15 example site is 'milton-todds-pit'. IF-15 lacks the backstop-radius rule. IF-02 lacks PARAM_REQUEST_READ re-requests. • ADD cites SRS D. Its drivers and cost table say £160/£150, and §2.1 says '£100 target'. §4.6 still says 'to confirm'. §4.8 says 'tool use'. The decision log lacks CR-04 and CR-05. It says 'ICD (next)'. • SRS: STK-05 still £100; the use story still has a key switch; the FS table caption says TBD-08; 'Next: ADD'. • SSS applicable documents are hard-coded to SRS D, ADD C and ICD B. HUL and REC are at Issue A with stale parents. • FMEA and KCL status/basis lines name superseded issues.	
Recommend	Create one baseline register in code (docs/common/baseline.py: document → issue → date). Have every build read its parents and references from it rather than from literals, and add a check that fails on a mismatch. Fix the listed text, then rebuild every PDF in one commit.	
Close with	Baseline register; all PDFs rebuilt; check passes	

RID-09	Major	Software baseline is not identifiable
Area	Configuration	Documents: SRS SWE-006; SSS-SIM SIM-D12; KCL; software/README

Finding	There are no git tags. There is no dependency lock file, so the SIM evidence cannot be reproduced exactly. The KCL archives ArduPilot sources at commit a11f7351, but the simulator is built from the Rover-4.7.1 branch (dbe79216). The evidence was therefore produced with a different ArduPilot commit from the one the documents cite.	
Recommend	Pin ArduPilot to one commit everywhere (KCL, README, build script) and re-archive the sources if needed. Add a lock file (pip-tools or uv). At freeze, create an annotated tag (e.g. sdr-baseline-1) that covers the documents, the parameters and the software, and record it in the baseline register.	
Close with	Lock file; single commit ID; tag	
RID-10	Minor	No CI, lint or type checking
Area	Configuration	Documents: SRS SWE-007 (Should); SSS-SIM SIM-D11; MCN-D33
Finding	No CI workflow exists. There is no ruff or mypy configuration. Unit tests and the secret-scan test run only when someone remembers.	
Recommend	Add a GitHub Actions workflow running ruff, mypy (on the boaty package) and the unit tests (SITL stays manual). Add a docs job that runs every build script's checks.	
Close with	Green workflow on the baseline tag	
RID-11	Minor	HDOP arming gate not configured
Area	Safety	Documents: SRS PRE-001; SSS-HLM HLM-D09; software/params
Finding	HLM-D09 says arming needs $\text{HDOP} \leq 1.5$, and allocates this to the helm. GPS_HDOP_GOOD is not in the parameter baseline , so ArduPilot's default (2.3) applies.	
Recommend	Add GPS_HDOP_GOOD = 150 to the baseline. Add a parameter read-back unit check.	
Close with	Parameter diff	
RID-12	Minor	Power budget file missing; cost margin thin
Area	Budgets	Documents: SSS-PWR PWR-D17; KCL cost reconciliation
Finding	PWR-D17 names docs/budgets/power.csv, which does not exist. The power analysis lives only in the KCL. The BOM is £181 against the £185 cap (margin £4, 2%) on estimated prices. The recovery-pole kit would breach it.	
Recommend	Either create power.csv from the KCL data (generated) or re-point PWR-D17 at the KCL section. Record an owner decision on what sits inside the cap (pole kit, spares), and state a price-margin policy for CDR (e.g. prices confirmed from vendors before ordering).	
Close with	File or SSS change; owner decision recorded	
RID-13	Minor	FMEA rule text and the build check disagree; missing failure mode
Area	Safety	Documents: FMEA §2.2 and fmea_data.check()
Finding	The rules say action is required for every $S \geq 9$ row 'whatever O and D are', but the check lets a direct test ($D \leq 3$) stand in for an action. SC-09 showed that re-arming mid-lake resets home to the re-arm point. VAL-004 catches it, but there is no failure mode for it.	
Recommend	Align the rule text with the check (state the $D \leq 3$ exemption). Add an FM for 'home reset by re-arm away from the launch point' with VAL-004 and SC-09 as controls.	
Close with	FMEA issue	
RID-14	Minor	Child's voice transcripts go to a cloud service without a stated policy
Area	Privacy	Documents: ICD IF-11; OPS manual; SRS NLI-008
Finding	Transcribed instructions from a 4-year-old are sent to the Claude API and logged locally. No document says what is sent, how long local logs are kept, or who may read them.	
Recommend	Add a short data statement to the ICD (IF-11 payload is text only, with no audio, names or location) and to the operations manual (log retention, how to delete).	
Close with	ICD and OPS text	
RID-15	Minor	Web UI security assumptions not recorded
Area	Security	Documents: ADD; ICD IF-09/IF-10; SSS-MCN

Finding	The web UI is plain HTTP. The panel and instruction endpoints rely on the closed Wi-Fi network, not on authentication (the adult PIN protects approval only). This is a reasonable choice for Mk1, but it is not written down as an assumption, and nothing requires WPA2 or a non-default passphrase.	
Recommend	Add the assumption to the ADD and a derived MCN requirement: WPA2/3 with a unique passphrase, the AP not bridged to the tethered internet for inbound traffic.	
Close with	ADD / SSS text	
OBS-01	Obs.	Site and permission open (TBD-02, TBD-03)
Area	Operations	Documents: SRS TBD register; site file
Finding	The site file is a stand-in built from public mapping. The real lake, launch point and home bay, and the Trust's permission, are open.	
Recommend	Carry. Gate: ORR (before the first lake trial). The site linter already blocks an unsurveyed site from use.	
Close with	Site survey; permission email	
OBS-02	Obs.	Hardware verification items open
Area	Verification	Documents: SSS-SIM V-01, V-07 to V-10; KCL
Finding	The rig and bench checks that close key assumptions (fence-aware RTL on hardware, ESC stop on signal loss, thrust, sag) are planned, but they need hardware.	
Recommend	Carry. Gate: CDR for the test plans, TRR for the results.	
Close with	Rig reports	
OBS-03	Obs.	ICD TBC items open
Area	Interfaces	Documents: ICD TBC-06, 08, 11-14
Finding	These need hardware or vendor data (pinouts, timings).	
Recommend	Carry. Gate: CDR.	
Close with	ICD issue at CDR	
OBS-04	Obs.	Performance numbers are estimates
Area	Budgets	Documents: KCL; SSS-PRP, PWR
Finding	Thrust, power draw, endurance and salvaged-cell capacity are estimates. Several datasheets are not yet fetched. The simulator boat model is tuned to these estimates, not measured data.	
Recommend	Carry. Gate: CDR (datasheets fetched, bench thrust and current measured, SITL model re-tuned to them).	
Close with	Bench data; model update	
OBS-05	Obs.	Planning latency not measured on the real link
Area	Performance	Documents: SRS NLI-007; NLI evaluation
Finding	The live evaluation met NLI-007 (p95 6.9 s) from a cloud container, not over a phone's 4G tether.	
Recommend	Carry. Gate: TRR (repeat SC-33 on the tethered Pi 5).	
Close with	SC-33 re-run record	
OBS-06	Obs.	Features not yet implemented
Area	Scope	Documents: SSS-MCN (MCN-D07, D51 and others)
Finding	Still to build: speech engines (STT and TTS), the fence editor, replay (MCN-D51), map tiles (MCN-D07), a waterfowl finder, RSSI and latency logging, and the QGroundControl takeover drill. These are specified, which is what PDR needs.	
Recommend	Carry. Gate: CDR (an implementation plan per item); TRR (built and verified).	
Close with	CDR plan	
OBS-07	Obs.	Remaining FMEA scenarios not built
Area	Verification	Documents: SSS-SIM SC-21 to SC-23, SC-34, SC-36, SC-39

Finding	These scenarios cover frozen GNSS, compass offset, helm restart, link degradation, interrupted upload and overstated capacity. Their FMEA detection ratings assume the tests will exist.
Recommend	Carry. Gate: TRR. None is needed to freeze the system design.
Close with	Pass records

7. Items carried to later gates

These are expected at SDR and do not block the freeze. Each has a gate by which it must close.

Item	Gate	What closes it
OBS-01 Site and permission open (TBD-02, TBD-03)	ORR (before the first lake trial). The site linter already blocks an unsurveyed site from use.	Site survey; permission email
OBS-02 Hardware verification items open	CDR for the test plans, TRR for the results.	Rig reports
OBS-03 ICD TBC items open	CDR.	ICD issue at CDR
OBS-04 Performance numbers are estimates	CDR (datasheets fetched, bench thrust and current measured, SITL model re-tuned to them).	Bench data; model update
OBS-05 Planning latency not measured on the real link	TRR (repeat SC-33 on the tethered Pi 5).	SC-33 re-run record
OBS-06 Features not yet implemented	CDR (an implementation plan per item); TRR (built and verified).	CDR plan
OBS-07 Remaining FMEA scenarios not built	TRR. None is needed to freeze the system design.	Pass records

8. Gap-closure plan and freeze

8.1 Work packages

The RIDs group naturally into five packages. They are ordered so that the requirement changes (which need owner decisions) start first, and the rebuild and tag come last.

WP	Content	RIDs	Owner input
WP1 Decisions	CR-06: NLI-003 and MOD-005 wording; wildlife stand-off, season and OPS-005 rewording; cost-cap scope (pole kit); FM-02 acceptance if needed	04, 06, 12, 03	Yes: approve the CR
WP2 Safety config	OA_TYPE Dijkstra's fence avoidance; GPS_HDOP_GOOD; parameter checks	01, 11	No
WP3 Scenarios	SC-27, SC-11, SC-12, SC-13, SC-20 (+ SC-08 or a deviation); FMEA re-rating; new FM	01, 02, 03, 13	Deviation for SC-08 if chosen
WP4 Interfaces & VCRM	Event vocabulary module and IF-04 table; IF-11 data statement; VCRM generated from pytest markers; security assumption	05, 07, 14, 15	No
WP5 Baseline	Baseline register; fix stale text; power.csv; ArduPilot pin; lock file; CI; rebuild all PDFs; tag	08, 09, 10, 12	Sign the freeze

8.2 Freeze (SDR exit) criteria

- Every Major and Minor RID is closed with its evidence, or dispositioned by the owner in writing.
- All document builds pass their checks, including the new baseline-consistency check and the VCRM check.
- Full SITL and unit suites pass on the tagged commit, with no new known gaps.
- Tag sdr-baseline-1 created. From then on, any change to a frozen document goes through a numbered CR with impact noted on the other documents.
- The owner signs the decision record (section 9).

8.3 RID log

RID	Sev.	Status	Closed by (commit / CR)	Date
RID-01	Major	Open		
RID-02	Major	Open		
RID-03	Major	Open		
RID-04	Major	Open		
RID-05	Major	Open		
RID-06	Major	Open		
RID-07	Major	Open		
RID-08	Major	Open		
RID-09	Major	Open		
RID-10	Minor	Open		
RID-11	Minor	Open		
RID-12	Minor	Open		
RID-13	Minor	Open		
RID-14	Minor	Open		
RID-15	Minor	Open		
OBS-01	Obs.	Carried		
OBS-02	Obs.	Carried		
OBS-03	Obs.	Carried		

RID	Sev.	Status	Closed by (commit / CR)	Date
OBS-04	Obs.	Carried		
OBS-05	Obs.	Carried		
OBS-06	Obs.	Carried		
OBS-07	Obs.	Carried		

8.4 What CDR will expect

- Hull and pod drawings with weighed-mass and buoyancy estimates; electronics box layout; wiring diagram.
- Software design description for the remaining features (OBS-06), and code review of the safety-relevant paths.
- Datasheets fetched; bench thrust and current measured; the SITL boat model re-tuned (OBS-04).
- Rig and bench test procedures (L1-L3, BENCH) with pass criteria traced through the VCRM.
- Confirmed prices and an order list within the cap.

9. Decision record

Item	Entry
Review	System Design Review / PDR, gate 1
Baseline reviewed	commit 88b1cd7
Reviewer recommendation	Conditional GO: close 9 Major and 6 Minor RIDs, then freeze and tag
Owner decision	☐ GO ☐ Conditional GO ☐ NO-GO
Conditions / notes	
Signed (owner)	
Date	

Appendix A. SIM-stage Must requirements without a named test

53 Must requirements due at the SIM stage are not named in the trace of any catalogue test. Many are verified by unit tests or by derived-requirement tests; the VCRM (RID-07) should show which, and add tests for the rest.

Requirement t	Pri.	Text (abridged)
MOD-001	M	The system shall implement exactly these boat modes: DISARMED, HOLD, MANUAL, AUTO and RTL, as defined in section 2.3.
MOD-004	M	The system shall refuse to arm unless every pre-arm check (PRE-001 to PRE-008) passes, and shall state the failing check in plain language at Mission ...
MOD-005	M	When an AUTO mission completes, the boat shall enter RTL. On reaching home (within the waypoint acceptance radius) it shall enter HOLD.
MOD-006	M	While armed, the operator shall be able to command HOLD, RTL, MANUAL or STOP from Mission Control at any time, overriding AUTO.
PRE-002	M	A geofence shall be loaded and enabled, and home shall lie inside the inclusion fence and outside all exclusion zones.
PRE-005	M	For AUTO: a mission that has passed validation and operator approval shall be loaded, and read back from the helm to match (VAL-010).
NAV-004	M	In AUTO the boat shall pass within the acceptance radius of each waypoint. The radius is configurable from 1 to 10 m, default 3 m.
NAV-007	M	In RTL the boat shall navigate to home without entering any exclusion zone or leaving the inclusion fence.
FEN-001	M	The system shall support one inclusion polygon of 3 to 70 vertices and up to 10 exclusion zones, each a polygon or circle.
FEN-002	M	Fences shall be drawn on a map at Mission Control, stored per site with a name, and version-controlled.
FEN-004	M	The fence shall be enforced in every armed mode, including MANUAL.
FEN-005	M	On a fence breach the boat shall enter RTL within 1 s of the breach.
FEN-006	M	If the boat stays outside the inclusion fence for more than 30 s, or gets more than 10 m outside it, the motors shall stop (HOLD, no thrust) and an al...
FEN-007	M	Fence and exclusion zones shall not be changed while the boat is armed.
SAF-002	M	Those functions shall keep working with the mission computer, Mission Control and the radio link all failed.
SAF-003	M	The mission computer and Mission Control shall command the helm only through the helm interface (SWE-002). They shall not be able to disable the fence...
SAF-004	M	Output from the language model shall never be sent to the helm as a command. Only mission objects that have passed validation (VAL-001) may reach the ...
SAF-006	M	A failure modes and effects analysis (FMEA) shall be done before the first lake trial and updated after any change to hardware or safety software.
MIS-001	M	The system shall support these mission elements: waypoint, area survey (lawnmower pattern over a polygon, with lane spacing), photo point (hold, then ...
MIS-002	M	Every mission shall end with return home.
MIS-003	M	Planned mission duration shall not exceed 20 min (configurable up to 30 min). Planned energy shall not exceed 50% of usable battery energy.
MIS-004	M	Missions shall be represented in a documented JSON schema with a version number.
MIS-005	M	Offline templates shall be available without internet, at least: 'Explore the bay', 'Duck patrol' (survey with photo points) and 'Lap of the bay'.
NLI-001	M	Mission Control shall accept instructions as typed text.
NLI-003	M	Instructions shall be turned into a mission through the Claude API. The request shall include the site context: fence, exclusion zones, home, battery ...

Requirement	Pri.	Text (abridged)
NLI-004	M	The language model's reply shall be constrained to the mission schema (structured output or tool use). A malformed reply shall be rejected and retried...
NLI-005	M	The proposed mission shall be shown on the map with a one-sentence plain-language summary, which shall also be spoken aloud.
NLI-006	M	Requests the boat must not or cannot do (for example leaving the fence, chasing animals, going to another lake) shall be declined in child-friendly wo...
NLI-008	M	The API key shall be stored only on Mission Control and shall never be committed to the repository.
VAL-001	M	Every mission, from any source, shall pass the validator before upload. There shall be no other path for uploading missions to the helm.
VAL-002	M	Every waypoint, and every leg sampled at 1 m intervals or finer, shall be inside the inclusion fence by at least 3 m and outside every exclusion zone ...
VAL-003	M	The mission shall meet the duration and energy limits in MIS-003.
VAL-004	M	The mission shall start within 10 m of home and end with return home.
VAL-005	M	Commanded speeds shall be within NAV-003. Each hold shall be 60 s or less. The photo count shall fit in the free camera storage. The waypoint count sh...
VAL-006	M	The validator shall be pure Python with no network access and no randomness. It shall have a unit-test suite that includes adversarial missions, with ...
VAL-007	M	Each rejection shall be reported in plain language, naming the rule broken and where on the map.
VAL-008	M	After validation, an adult shall approve the mission at Mission Control. GO shall stay disabled until then.
VAL-009	M	Any change to an approved mission shall void the approval.
DET-004	M	If DET-003 is implemented, 'pause and look' shall hold position and shall never steer towards the detected animal.
MC-003	M	STOP and COME HOME shall always be active while armed. GO shall be active only when a validated, approved and verified mission is loaded and the boat ...
MC-005	M	The display shall show, updated at least once a second: a map with the boat's position, heading and track; fence, exclusions, home and mission; batter...
MC-009	M	The last known position and track shall stay displayed after link loss, in a 'find my boat' view.
MC-010	M	Mission Control shall host the pre-launch checklist (OPS-001) and record who signed it off, and when.
MC-011	M	Everything except natural-language planning and cloud photo analysis shall work without internet access.
SWE-001	M	All project-written software shall be Python: CPython 3.11+ on Linux and MicroPython on microcontrollers. Third-party firmware (ArduPilot, ESC firmwar...
SWE-002	M	A Python helm interface shall provide: arm, disarm, set mode, upload mission, read mission back, start, hold, return home, stop, set fence, and a stat...
SWE-003	M	Mission-level code shall run unchanged against any helm implementation, selected by configuration: first ArduPilot over MAVLink, later a Python helm.
SWE-005	M	Each failsafe requirement (FS-001 to FS-013) shall have an automated simulation scenario.
SWE-006	M	Source, configuration and safety parameters shall live in this git repository, with a tagged release for each lake trial.
LOG-002	M	Mission Control shall log: all commands, sign-offs and approvals; each mission as approved (JSON); language model requests and replies; and validator ...
LOG-003	M	The logs shall be enough to replay any trip on a map, including every failsafe event.
LOG-004	M	Photos shall be stored locally and never published automatically. Photos go to cloud services only if an adult enables cloud analysis for that session...
OPS-012	M	Before the first lake trial, the operator shall rehearse every contingency scenario in section 3.5 in simulation.